

Problem 16.8

An annuity pays \$500 every six months for 5 years. The annual rate of interest is 8% convertible semiannually. Find each of the following:

- (a) the present value of the annuity six months, one period, before the first payment,
- (b) the present value of the annuity on the day of the first payment,
- (c) the future value of the annuity on the day of the last payment,
- (d) the future value of the annuity six months after the last payment.

Solution

The annual nominal interest rate convertible semiannually is 8%, so the six-month effective interest rate is

$$j = \frac{0.08}{2} = 0.04.$$

There are

$$5(2) = 10$$

semiannual payments.

Part (a)

Six months before the first payment, the payments form an annuity-immediate. Thus,

$$PV = 500a_{\overline{10}|j}.$$

BAII Plus:	
Key	Value
2ND + BGN	END mode
2ND + QUIT	return
2ND + CLR WORK	clear TVM
N	10
I/Y	4
PMT	500
FV	0
CPT PV	-4055.45

Using

$$a_{\overline{n}|j} = \frac{1 - (1 + j)^{-n}}{j},$$

we get

$$\begin{aligned} PV &= 500 \left(\frac{1 - (1.04)^{-10}}{0.04} \right) \\ &\approx 4055.4475. \end{aligned}$$

Part (b)

On the day of the first payment, the value is one period later than part (a), so

$$PV = 500\ddot{a}_{\overline{10}|}.$$

BAII Plus:	
Key	Value
2ND + BGN	BGN mode
2ND + SET	set beginning
2ND + QUIT	return
2ND + CLR WORK	clear TVM
N	10
I/Y	4
PMT	500
FV	0
CPT PV	-4217.67

Therefore,

$$\begin{aligned} PV &= 500(1.04) \left(\frac{1 - (1.04)^{-10}}{0.04} \right) \\ &\approx 4217.6654. \end{aligned}$$

Part (c)

On the day of the last payment, the future value is

$$FV = 500s_{\overline{10}|}.$$

BAII Plus:	
Key	Value
2ND + BGN	END mode
2ND + QUIT	return
2ND + CLR WORK	clear TVM
N	10
I/Y	4
PMT	500
PV	0
CPT FV	6003.05

Using

$$s_{\overline{n}|j} = \frac{(1+j)^n - 1}{j},$$

we get

$$FV = 500 \left(\frac{(1.04)^{10} - 1}{0.04} \right) \\ \approx 6003.0532.$$

Part (d)

Six months after the last payment, the value is one period after part (c), so

$$FV = 500s_{\overline{10}|4}.$$

BAII Plus:	
Key	Value
2ND + BGN	BGN mode
2ND + SET	set beginning
2ND + QUIT	return
2ND + CLR WORK	clear TVM
N	10
I/Y	4
PMT	500
PV	0
CPT FV	6243.18

Therefore,

$$FV = 500(1.04) \left(\frac{(1.04)^{10} - 1}{0.04} \right) \\ \approx 6243.1753.$$

Hence,

- (a) 4055.45,
- (b) 4217.67,
- (c) 6003.05,
- (d) 6243.18.